# MATH MA’AM ACADEMY

Where Every Subject Clicks.

DR. HOMI BHABHA BALVAIDNYANIK COMPETITION

STANDARD VI • SCIENCE

# UNIT 2: CLASSIFICATION OF ANIMALS

Revision notes based on the supplied textbook pages, expanded with essential classification concepts and examples for Homi Bhabha-level preparation.

## QUICK CLASSIFICATION MAP

| Basis | Main groups |
|---|---|
| Animal kingdom | Non-chordates and chordates |
| Non-chordate phyla | Porifera, Coelenterata (Cnidaria), Platyhelminthes, Aschelminthes (Nematoda), Annelida, Arthropoda, Mollusca, Echinodermata |
| Vertebrate classes | Pisces, Amphibia, Reptilia, Aves, Mammalia |
| Habitat | Terrestrial and aquatic; terrestrial includes desert, grassland, rainforest, tundra/polar and mountain habitats |
| Food habit | Autotrophs (producers) and heterotrophs (consumers); consumers include herbivores, carnivores and omnivores |

# 1. BASIC IDEAS IN ANIMAL CLASSIFICATION

- Classification: systematic grouping of organisms based on similarities, differences and evolutionary relationships.
- Kingdom Animalia: animals are multicellular, eukaryotic and heterotrophic. Most can move at least during some stage of life; their cells do not have cell walls.
- Invertebrates: animals without a vertebral column (backbone). Many belong to the non-chordate groups listed in this unit.
- Vertebrates: animals with a vertebral column and an internal skeleton made of bone or cartilage.
- Chordates: animals that possess a notochord at least during some stage of development. Vertebrates are a major group within chordates.
- Symmetry: arrangement of body parts around an axis or plane. Common types are asymmetry, radial symmetry and bilateral symmetry.

## Body symmetry

| Type | Meaning | Examples |
|---|---|---|
| Asymmetrical | Body cannot be divided into equal halves through a regular plane. | Many sponges |
| Radial | Parts are arranged around a central axis; several planes can produce similar halves. | Hydra, jellyfish, adult starfish |
| Bilateral | Only one main plane divides the body into right and left mirror-image halves. | Earthworm, insect, fish, human |

# 2. NON-CHORDATE ANIMALS

The following phyla are commonly studied in this unit. Habitat and examples can vary within a phylum; the identifying body feature is especially useful for classification.

| Phylum | Main features | Examples |
|---|---|---|
| Porifera | Mostly aquatic; body has numerous pores; usually asymmetrical; water passes through a canal system. Sessile adults; cellular level of organisation. | Sycon, Euspongia (bath sponge), Spongilla (freshwater sponge), Euplectella |
| Coelenterata / Cnidaria | Mostly aquatic, mainly marine; radial symmetry; tentacles around the mouth; stinging cells called cnidocytes. | Hydra, sea anemone, jellyfish, coral, Physalia (Portuguese man-of-war) |
| Platyhelminthes | Flatworms; dorsoventrally flattened, soft, unsegmented body; bilateral symmetry; many are parasitic. | Planaria, tapeworm, liver fluke |
| Aschelminthes / Nematoda | Roundworms; long, cylindrical, unsegmented body; bilateral symmetry; many are parasitic, some free-living. | Ascaris, Wuchereria (filarial worm), Loa loa |
| Annelida | Segmented worms; bilateral symmetry; body divided into ring-like segments; many have bristles (setae). | Earthworm, Nereis, leech |
| Arthropoda | Largest animal phylum; jointed appendages, segmented body and chitinous exoskeleton; bilateral symmetry. | Crab, prawn, butterfly, housefly, spider, scorpion, centipede |
| Mollusca | Soft, usually unsegmented body; muscular foot in many; mantle often secretes a calcareous shell; bilateral symmetry in basic body plan. | Oyster, Pila, snail, squid, octopus, cuttlefish, shipworm |
| Echinodermata | Exclusively marine; spiny skin; water vascular system; adults usually show five-part radial symmetry, larvae bilateral. | Starfish, sea cucumber, brittle star, sea urchin |

### High-value distinctions

- Sponges have pores; cnidarians have tentacles and stinging cells.
- Flatworms are flattened; roundworms are cylindrical and unsegmented; annelids are segmented.
- Arthropods have jointed appendages and a chitinous exoskeleton.
- Molluscs are soft-bodied; a shell is common but not present in every mollusc (for example, octopus has no external shell).
- Echinoderms are marine only. Adult starfish show radial symmetry, while their larvae are bilateral.

# 3. VERTEBRATES: FIVE MAJOR CLASSES

| Feature | Pisces | Amphibia | Reptilia | Aves | Mammalia |
|---|---|---|---|---|---|
| Habitat | Aquatic | Land and water | Mainly land; some aquatic | Terrestrial; some aerial/aquatic | Mainly land; some aquatic/aerial |
| Body covering | Scales | Moist, usually scaleless skin | Dry scales or scutes | Feathers | Hair, fur or wool |
| Body plan | Usually streamlined; fins | Head and trunk; usually two pairs of limbs in adults | Head, neck, trunk; usually two pairs of limbs (snakes lack limbs) | Beak; forelimbs modified into wings | Head, neck, trunk; usually four limbs |
| Respiration | Gills | Larvae usually gills; adults lungs and skin | Lungs | Lungs with air sacs | Lungs; diaphragm in most |
| Reproduction | Mostly oviparous; usually eggs in water | Usually eggs in water; metamorphosis common | Mostly oviparous; some are viviparous | Oviparous; hard-shelled eggs | Mostly viviparous; monotremes lay eggs |
| Temperature | Ectothermic | Ectothermic | Ectothermic | Endothermic | Endothermic |
| Examples | Shark, rohu, seahorse, electric ray | Frog, toad, salamander | Tortoise, lizard, snake, crocodile | Peacock, parrot, crow, duck, penguin | Kangaroo, dolphin, bat, dog, cat, human |

### Remember

- Endothermic: body temperature is regulated largely by internal metabolic heat (birds and mammals). Ectothermic: body temperature depends more on external conditions (fish, amphibians and reptiles). 'Warm-blooded' and 'cold-blooded' are common school terms, but endothermic/ectothermic are more precise.
- Not all fish have a typical streamlined body; not all reptiles have four limbs; not all mammals give birth to live young.

# 4. IMPORTANT VERTEBRATE FACTS

- Pisces: aquatic vertebrates; most breathe through gills and move with fins. Many have a swim bladder for buoyancy, but sharks generally do not.
- Amphibia: commonly have a life cycle involving water and land. Frog tadpoles breathe through gills; adult frogs use lungs and moist skin. Metamorphosis changes a larva into an adult.
- Reptilia: dry, scaly skin reduces water loss; respiration occurs through lungs. Most lay leathery-shelled eggs on land, though some species give birth to live young.
- Aves: feathers are unique to birds; forelimbs are modified into wings. Birds have lungs connected to air sacs and are endothermic. Penguins are birds that cannot fly.
- Mammalia: mammary glands produce milk to feed young; hair or fur is characteristic. Most are viviparous, but platypus and echidnas are egg-laying mammals.

## Special examples that often appear in MCQs

| Animal | Correct group / special fact |
|---|---|
| Whale and dolphin | Mammals; breathe with lungs and feed young with milk, despite living in water. |
| Bat | Mammal; forelimbs are modified into wings. |
| Penguin | Bird; has feathers and wings but cannot fly. |
| Flying fish | Fish; glides above water but does not become a bird. |
| Shark | Cartilaginous fish; skeleton is mainly cartilage. |
| Seahorse | Fish; male carries developing eggs in a brood pouch. |
| Crocodile | Reptile; has a four-chambered heart among living reptiles. |
| Platypus | Egg-laying mammal (monotreme). |
| Frog | Amphibian; undergoes metamorphosis from tadpole to adult. |

# 5. CLASSIFICATION BASED ON HABITAT

A habitat is the natural place or environment in which an organism lives. Animals may be grouped broadly as terrestrial or aquatic.

| Habitat type | Description | Examples |
|---|---|---|
| Desert | Very low rainfall; organisms show adaptations to conserve water and tolerate heat or cold. | Camel, desert fox, lizard |
| Grassland | Dominated by grasses; supports grazing herbivores and their predators. | Giraffe, zebra, antelope, lion |
| Rainforest | Warm, often wet forests with dense vegetation and many ecological niches. | Tiger, snakes, monkeys, many birds |
| Tundra / polar | Very cold regions with short growing seasons; polar habitats may have ice and snow. | Polar bear, reindeer, penguin (Antarctic) |
| Mountain | High-altitude habitats with lower temperatures and thinner air. | Yak, mountain goat, snow leopard |
| Freshwater | Rivers, lakes, ponds and streams with low salt concentration. | Rohu, frog, freshwater mussels |
| Marine | Seas and oceans with saline water. | Shark, whale, octopus, starfish |

## Habitat adaptations

- Camels conserve water and can tolerate large changes in body-water balance; broad feet help them walk on sand.
- Polar animals often have insulation such as thick fur, feathers or fat (blubber).
- Aquatic animals may have streamlined bodies and fins or flippers for movement through water.
- Mountain animals may have adaptations for cold and low oxygen availability.

# 6. CLASSIFICATION BASED ON FOOD HABITS

Food habits describe how organisms obtain energy and nutrients. Green plants are producers; animals depend directly or indirectly on other organisms for food.

| Group | Meaning | Examples |
|---|---|---|
| Autotrophs / producers | Make their own organic food, usually by photosynthesis. | Green plants, algae |
| Herbivores | Animals that mainly eat plants or plant parts. | Cow, deer, rabbit, giraffe |
| Carnivores | Animals that mainly eat other animals. | Lion, tiger, wolf |
| Omnivores | Animals that eat both plant and animal matter. | Crow, bear, human |
| Scavengers | Feed on dead animals; help remove carcasses. | Vulture, hyena, jackal, crow |
| Decomposers | Break down dead organic matter and wastes into simpler substances; recycle nutrients. | Bacteria, fungi |

## Important distinctions

- Scavengers feed on dead organisms; decomposers chemically break down organic matter. Some animals, such as hyenas, may both hunt and scavenge.
- A food chain shows one pathway of energy transfer. A food web is a network of interconnected food chains.
- Energy generally flows from producers to consumers and decomposers; nutrients are recycled through ecosystems.

# 7. SPECIALISATION OF TEETH ACCORDING TO FOOD HABIT

| Type of teeth | Main function | Common examples |
|---|---|---|
| Incisors | Chisel or cut food; located at the front. | Well developed in many herbivores for cutting plants. |
| Canines | Pierce, grip and tear food. | Usually prominent in carnivores. |
| Premolars and molars | Crush and grind food. | Broad grinding surfaces in many herbivores and omnivores. |

- Herbivores generally have broad incisors for cutting vegetation and broad, ridged premolars/molars for grinding. Canines may be reduced or absent in many herbivores.
- Carnivores generally have sharp canines for tearing and blade-like cheek teeth for slicing flesh.
- Omnivores have a combination of cutting, tearing and grinding teeth.

# 8. SCIENTIFIC TERMS TO KNOW

| Term | Definition |
|---|---|
| Vertebral column | A series of vertebrae forming the backbone. |
| Notochord | A flexible supporting rod present in chordates at least during embryonic development. |
| Exoskeleton | A hard external supporting or protective covering. |
| Endoskeleton | An internal supporting framework, made of bone or cartilage in vertebrates. |
| Oviparous | Animals that lay eggs. |
| Viviparous | Animals that give birth to live young. |
| Metamorphosis | A major change in body form during development, such as tadpole to frog. |
| Ectothermic | Body temperature depends substantially on external environmental heat. |
| Endothermic | Uses internal metabolic processes to maintain a relatively stable body temperature. |
| Bilateral symmetry | One plane divides the body into similar right and left halves. |
| Radial symmetry | Body parts are arranged around a central axis. |
| Segmentation | Division of the body into repeated sections or segments. |

## Final revision strategy

- Learn the diagnostic feature of each non-chordate phylum and match it with one or two standard examples.
- Compare the five vertebrate classes by body covering, respiration, reproduction and temperature regulation.
- Practise unusual examples: whale, bat, penguin, platypus, shark, seahorse and flying fish.
- Do not classify an animal by habitat alone: whales are mammals, penguins are birds and dolphins are mammals.

Understand the identifying features, comparisons, examples and exceptions—not just the names of the groups.